nature portfolio

Peer Review File

Artificial Intelligence Links CT Images to Pathologic Features and Survival Outcomes of Renal Masses

Editorial Note: This manuscript has been previously reviewed at another journal that is not operating a transparent peer review scheme. This document only contains reviewer comments and rebuttal letters for versions considered at *Nature Communications*.

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

Thank you for making the code available for the readers. However, it me be helpful to share the trained weights as well.

Reviewer #1 (Remarks on code availability):

I checked the code in the repository, which seems reasonable and can be used, but didn't run it since the data/pre-trained model is not available.

Reviewer #4 (Remarks to the Author): Expert in AI, machine learning, and computer vision; replaces Reviewer #2

The authors developed deep learning based diagnostic models for classifying renal masses and renal tumour. Particularly, CT images collected from multi centres were used to validate the classification performance, and the experimental results showed that the proposed models outperformed the radiomics-based model. Besides, they showed the potential association of AI predictions with survival and gene mutation. Overall. the methodology is sound, and the experiments can support the conclusions and claims, but I still have some concerns.

Major comments:
1. Interpretation of predictions. This paper developed several deep learning models based the different combinations of CT phases. Table s6 shows that the three-phase-based model achieves the best classification perform, and the venous phase-based model outperforms the non-contrast and arterial phase-based models. However, for the multi-phase classification model, the authors have not shown what does each phase contribute to the final predictions. Moreover, which kinds of radiological phenotypes in each phase are highly correlated to the predictions.
2. Annotation of data. In section of histologic classification, the authors have not reported the intra-observer variability of the three pathologists in revaluating tumour subtypes. The arterial phase CT images for fine-tuning the segmentation model was only labelled by one radiologist, this may lead to annotation bias.
3. It is still unconvincing why the authors only extract one slice with maximum tumour for each CT image based on the tumour segmentation, and no experiment has been provided to show its strength. Spatial

heterogeneity is a fundamental biological feature of tumour. Only using one slice may ignore the important spatial heterogeneity. Besides, this strategy may raise new issues on model generalization to small tumours. I recommend the authors extract the other slice with a smaller tumour for each CT image to test the model generalizability. Moreover, the nnUnet for tumour segmentation may be failed for some difficult images from TCIA. How did you deal with this case?
4. After the segmentation, why not focus on kidney areas by masking background, but crop volumes including much background information which may mislead diagnosis? For example, in Figure 2, although the aim is to discriminate the benign or malignancy of renal mass, but the benign CAM in Figure C shows that much attention has been paid to background areas. Alternatively, why not include segmentation results as additional information for classification?
5. For the feature extractor ResNet18, the authors have never mentioned if they trained or finetuned ResNet18 using CT images, so I suppose they directly used the ResNet18 trained on ImageNet. In this case, how to ensure the ResNet18 trained using natural images could generalize well to medical images and particularly extract representative features. Why not use the nnUnet trained using CT images for segmentation to extract features? What if the multi-scale features extracted by the nnUnet from cropped 3D volumes are used for binary classification? In Table s3, all backbones were trained on ImageNet, but tested on CT slices. Due to the distribution shift between training and test, it is unfair to compare them. The authors should also include backbones which have met CT images for comparison.
6. AUC cannot completely represent the classification performance, especially when samples are imbalanced, which however is the case for this work. The authors also need to report AUPRC, which is much better for imbalanced data. In table s4, although the classification model achieves comparable performance in terms of AUC, but significantly lower accuracy and sensitivity scores for TCIA, demonstrating higher false negative rate during diagnosis and leading to loss of effective treatments for patients. One reason may be the portion of benign cases in TCIA is only 3.6%, which is greatly smaller than that of training set (16.8%). However, my main concern is about the race bias of this AI model. The patients of training set completely come from China, while most of patients in TCIA are white. The authors should include the statistics of race in table 1 and report the classification performance for each race group to validate if their models exist the issue of race bias.

Minor comments:
1. Line 107-109, supplementary material, for patients without arterial phase images in TCIA dataset, how to achieve the registration to the arterial phase image?
2. For binary classification task, although 2000+ patients were included in training set, the total number of training images are still limited since only one slice was extracted from each CT images. In this case, it is better to train the classifiers by cross-validation.
3. Line 505, why "survival outcome is the gold standard for aggressiveness evaluation"? Survival outcomes are end points. In fact, Gleason score of biopsy samples is the widely used to evaluate the aggressiveness of cancer.
4. Line 632-634, model overfitting occurs due to limited training samples, not limited test samples. Limited test samples could raise the risk of model generalization as well as model bias.
5. Line 649-650, how did you include both tumour nucleus grade and histologic subtype in predicting aggressiveness of renal tumours? Figure shows that only CT images were fed into deep learning models.
6. Line 664, an AUC of 0.619 seems low. In general, the AUC of a random classifier is 0.5, so it seems that the diagnostic model cannot distinguish well between indolent and aggressive tumours in this study.

7. The information of spatial resolution for CT scans has not been provided.

Reviewer #4 (Remarks on code availability):

The authors used the source code of nnUnet for kidney and tumor segmentation, which is reproducible. They used pretrained ResNet on ImageNet to extract features, which is reproducible. Besides, they provided the source code of classification, which is reproducible. Overall, the provided README file is readable, and the source code is a usable resource for the community.

Reviewer #5 (Remarks to the Author): Expert in radiology and kidney cancers; replaces Reviewer #3

Introduction: Acceptable

Methods:
-agree with authors to classify histology as defined rather than based on stage and size.
-for definition of indolent lesions under histological section, did the authors mean to say invasion instead of evasion? - "included indolent ccRCC (ccRCC without evasion into major veins or perinephric tissues, grade 3-4 components, necrosis or sarcomatoid differentiation), indolent papillary RCC (papillary RCC without evasion into major veins or perinephric tissues, grade 3-4 components or sarcomatoid differentiation)"
-adding the subset analysis for T1/T1a tumors is acceptable
-agree with authors regarding difficulty to control for size having more aggressive features
- added section on radiological interpretation is acceptable. I would also add if the 7 readers/expert had any fellowship training in abdominal imaging as part of their 5 years of experience. i.e. Each were fellowship trained in abdominal imaging with over 5 years of experience...
-No other concerns in methods section

Results:
Line 436 - change than to then

Discussion:
Line 734- typo for "actual"

Table S5 Title - typo errors "radiologists" and "fixed"

Table s6 typo: AUCs for deep "learnig" model x 2

Overall summary: Excellent manuscript for which the authors adequately addressed the reviewer's concerns.

Reviewer #5 (Remarks on code availability):

NA

Point by point responses to reviewers

Reviewer #1 (Remarks to the Author):

Thank you for making the code available for the readers. However, it me be helpful to share the trained weights as well.

Response: We appreciate the reviewer's suggestion. To facilitate further research and reproducibility, we will added instructions for obtaining the trained model weights (https://drive.google.com/drive/folders/1jrvlf16Mi01VhjsT2aoKS9cOtiPkjU0T) in the GitHub repository.

Reviewer #1 (Remarks on code availability):

I checked the code in the repository, which seems reasonable and can be used, but didn't run it since the data/pre-trained model is not available.

Response: Thank you for taking the time to check the code. Following your suggestions, we make the pre-trained model weights available to help readers reproduce the findings from our study.

Reviewer #4 (Remarks to the Author):

The authors developed deep learning based diagnostic models for classifying renal masses and renal tumour. Particularly, CT images collected from multi centres were used to validate the classification performance, and the experimental results showed that the proposed models outperformed the radiomics-based model. Besides, they showed the potential association of AI predictions with survival and gene mutation. Overall, the methodology is sound, and the experiments can support the conclusions and claims, but I still have some concerns.

Response: Thank you for your thoughtful review and for recognizing the strengths of our work. We appreciate your valuable feedback, which has helped us to improve the clarity and rigor of the manuscript. We have carefully revised the manuscript to address your specific concerns, as detailed in the responses below. Additionally, as some similar issues (e.g., 3D vs. 2D technical consideration, quality control of automatic segmentation) were raised and addressed during the first-round review (which might be not visible in last transferred submission to *Nature Communication*), we have attached our responses to those comments for your reference to ensure full transparency.

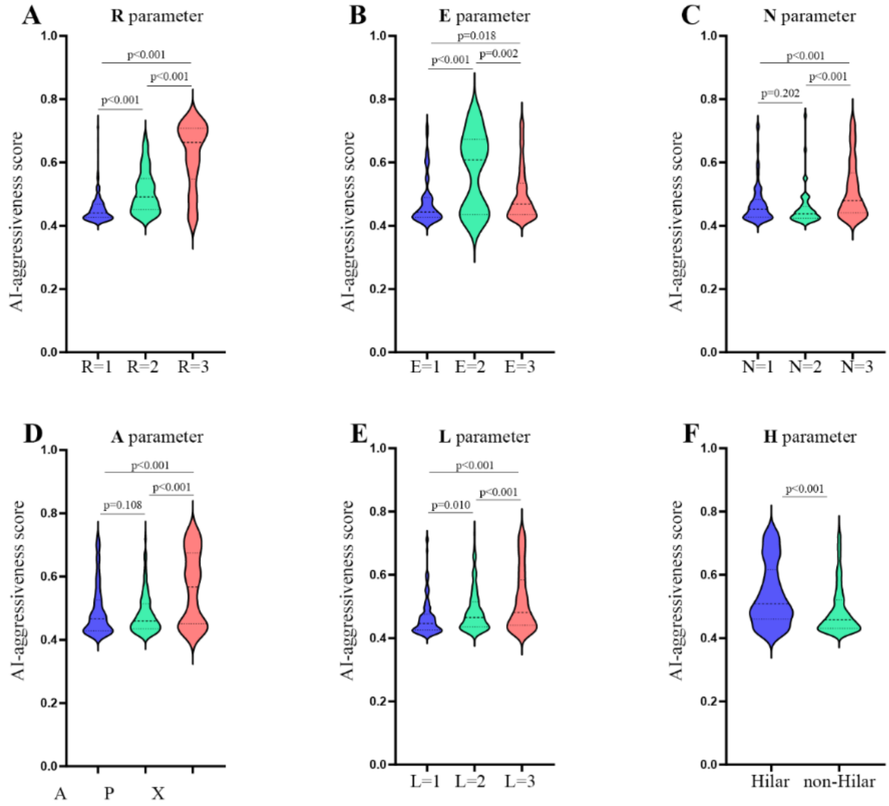
Major comments:

1. Interpretation of predictions. This paper developed several deep learning models based the different combinations of CT phases. Table S6 shows that the three-phase-based model achieves the best classification perform, and the venous phase-based model outperforms the non-contrast and arterial phase-based models. However, for the multi-phase classification model, the authors have not shown what does each phase contribute to the final predictions. Moreover, which kinds of radiological phenotypes in each phase are highly correlated to the predictions.

Response: Thank you for your insightful comments. As you mentioned, our comparison of multi-phase, dual-phase, and single-phase models demonstrated the additional benefits of integrating complementary phases (Table S6). In Figure 2C and Figure 3C, we employed the Class Activation Mapping (CAM) technique to visualize the saliency map. However, CAM often produces coarse-resolution maps and offers limited explainability [Ref 1]. This limitation constrains its ability to provide precise insights into the contribution of each CT phase to the final predictions.

To further address your concern, we analyzed the associations between the R.E.N.A.L. nephrometry score [Ref 2], one of the most established radiological classification systems for renal tumors, and the AI model's predictions of tumor aggressiveness in the external test set. The R.E.N.A.L. score is based on five key anatomical features: R (radius, the maximum diameter of the tumor), E (exophytic/endophytic properties), N (nearness to the collecting system or sinus), A (anterior/posterior location), and L (location relative to the kidney's polar line), with the suffix 'h' indicating contact with the renal artery or vein. Although radiological classification systems like R.E.N.A.L. are not phase-specific and primarily assess anatomical features, we found significant differences in AI-aggressiveness scores across different levels of each R.E.N.A.L. parameter (Figure R: A~F). This suggests a strong correlation between the R.E.N.A.L. score and our model's predictions for aggressiveness.

Figure R



In response to your request to clarify the contribution of each phase, we have investigated the selected features in the radiomics model, where the features have explicit definitions. Our analysis demonstrated that Cluster Prominence, a texture feature measuring local intensity variation, played a significant role in malignancy prediction across the non-contrast (N), arterial (A), and venous (V) phases. Additionally, for predicting tumor invasiveness, features such as tumor size and sphericity were critical, along with certain first-order histogram features that describe the distribution of voxel intensities within the tumor region.

Radiomics Model	**Selected Features**
Maligancy prediction	V_glcm_ClusterProminence
	V_glcm_ClusterShade
	A_glcm_ClusterProminence
	V_ngtdm_Complexity
	A_firstorder_RootMeanSquared
	N_glcm_ClusterProminence
Invasiveness prediction	shape_MaximumDiameter
	N_firstorder_90Percentile
	A_firstorder_10Percentile
	A_firstorder_Kurtosis
	V_glszm_SmallAreaEmphasis
	V_firstorder_RootMeanSquared
	A_ngtdm_Coarseness
	A_firstorder_InterquartileRange
	A_glszm_SmallAreaEmphasis
	A_ngtdm_Busyness
	shape_Sphericity
	N_firstorder_Kurtosis
	V_glszm_SizeZoneNonUniformityNormalized
	V_gldm_GrayLevelNonUniformity

Reference:

[1] Ayhan, Murat Seçkin, et al. "Clinical validation of saliency maps for understanding deep neural networks in ophthalmology." Medical Image Analysis (2022).

[2] Kutikov, Alexander, and Robert G. Uzzo. "The RENAL nephrometry score: a comprehensive standardized system for quantitating renal tumor size, location and depth." The Journal of Urology (2009).

2.1. Annotation of data. In section of histologic classification, the authors have not reported the intra-observer variability of the three pathologists in revaluating tumour subtypes.

Response: Thank you for raising this point. Due to the large number of cases in our study, re-evaluating the histologic subtypes and tumor grades for each slide would have been a highly time-consuming task. To ensure efficiency and accuracy, each case was re-evaluated by one experienced pathologist from a tertiary medical center. In this scenario, intra-observer variability was not applicable, as each case was reviewed by a single pathologist rather than multiple pathologists reviewing the same cases. We focused on ensuring consistency by involving only highly experienced pathologists in the re-evaluation process.

2.2. The arterial phase CT images for fine-tuning the segmentation model was only labelled by one radiologist, this may lead to annotation bias.

Response: Thank you for your comment. Indeed, the 100 cases used for fine-tuning the nnUNet were labeled by one experienced radiologist (C.D.) following the KiTS dataset annotating protocol. It is important to note that the primary objective of the segmentation was to locate the kidney and tumor center, rather than achieve perfect boundary delineation (which will be further discussed in the next response).

We regret any confusion caused by the omission of our quality control process. To mitigate potential errors in location, we implemented a quality control step wherein a radiologist reviewed the segmentation results. In instances where the automatic segmentation was suboptimal, the radiologist manually placed a mark on the tumor center in the maximal slice using the ‘paint’ tool in 3D Slicer, thereby replacing the automatically generated results. The quality assured center ensures that the correct region of interest (ROI) was cropped for downstream prediction. The quality control detail was consistent with our recent work [Ref 1].

Reference:

[1] Dai, Chenchen, et al. "Deep Learning Assessment of Small Renal Masses at Contrast-enhanced Multiphase CT." Radiology (2024).

3.1. It is still unconvincing why the authors only extract one slice with maximum tumour for each CT image based on the tumour segmentation, and no experiment has been provided to show its strength. Spatial heterogeneity is a fundamental biological feature of tumour. Only using one slice may ignore the important spatial heterogeneity. Besides, this strategy may raise new issues on model generalization to small tumours. I recommend the authors extract the other slice with a smaller tumour for each CT image to test the model generalizability.

Response: Thank you for your insightful comments. Regarding the use of a 2D cropped slice instead of 3D images, we would like to reference our response to Reviewer #1 from the first-round review:

" How was 2D (Res-18) to 3D conversion was handled, what region of interest were used? Any pre-processing was applied?

Response: Thank you for highlighting the technical aspects. Detailed information regarding the network configuration and training specifics is available in the Supplementary Materials. Our approach involved using a segmentation network to identify masses before processing the images through the multi-phase network. Specifically, we focused on the region of interest that contains renal tumors, cropping it accordingly. For preprocessing, we selected axial slices showcasing the largest tumor dimension and resampled them to a spatial resolution of 0.625×0.625 mm for 2D network input. This methodology was deliberately chosen to mitigate overfitting risks. We opted for a 2D slice representing the maximum tumor area and incorporated a ResNet-18 backbone, aiming to balance complexity and performance. While more sophisticated 3D networks, such as Vision Transformers, possess greater learning capabilities, our testing indicated a tendency for these models to overfit, leading to suboptimal performance on test datasets. After conducting extensive experiments across various network architectures, we concluded that a simpler design, coupled with effective regularization strategies, was the most suitable approach for our objectives."

We recognize that using only the maximal slice representing the tumor sacrifices some degree of spatial heterogeneity. However, this 2D strategy significantly reduces the risk of overfitting, which is a key concern when dealing with limited medical imaging data. Additionally, the varying slice thicknesses present challenges for 3D convolutional operations, making the 2D approach more efficient and effective. This approach has also been successfully employed in similar studies [Ref 1-2], and its effectiveness on small renal tumors has been validated in our concurrent work [Ref 3].

Following your suggestion, we conducted additional experiments by randomly selecting a slice from the three nearest slices in the internal test sets:

	AUC for malignancy prediction	AUC for invasiveness prediction
Maximum tumor slice	0.898	0.792
Radom slice from 3 nearest	0.875	0.771

The results showed only a slight decrease in classification performance, demonstrating the robustness of the model to variations in slice selection. Additionally, we have provided detailed information about slice thickness in response to another of your comments. We agree that relying on a single maximal slice may limit the model's ability to capture the full 3D spatial heterogeneity, and we have added this point to the discussion section of the revised manuscript.

Reference:

[1] Uhm, Kwang-Hyun, et al. "Deep learning for end-to-end kidney cancer diagnosis on multi-phase abdominal computed tomography." NPJ precision oncology (2021).

[2] Xi, Ianto Lin, et al. "Deep learning to distinguish benign from malignant renal lesions based on routine MR imaging." Clinical Cancer Research(2020).

[3] Dai, Chenchen, et al. "Deep Learning Assessment of Small Renal Masses at Contrast-enhanced Multiphase CT." Radiology (2024).

3.2. Moreover, the nnUnet for tumour segmentation may be failed for some difficult images from TCIA. How did you deal with this case?

Response: Thank you for your comment. As detailed in the "Multi-Phase Convolutional Neural Network for Renal Mass Classification" section of the supplementary materials, we described how to address potential segmentation failures by implementing a quality control process. In cases where the nnUNet failed to accurately segment the tumor, an experienced radiologist (C.D.) manually reviewed the volumes and corrected the tumor center. This procedure ensured that the correct region of interest was identified for downstream analysis.

4. After the segmentation, why not focus on kidney areas by masking background, but crop volumes including much background information which may mislead diagnosis? For example, in Figure 2, although the aim is to discriminate the benign or malignancy of renal mass, but the benign CAM in Figure C shows that much attention has been paid to background areas. Alternatively, why not include segmentation results as additional information for classification?

Response: Thank you for your thoughtful comment. We chose not to focus exclusively on the kidney areas by masking the background because certain peri-tumor features in the surrounding tissues provide critical information for diagnosis. The aggressiveness of renal tumors is not solely determined by the tumor itself; important factors such as peripheral fat invasion, tumor thrombus, and adjacent tissue invasion—indicative of malignancy—often reside in the surrounding areas. By including this background information, we aim to capture these aggressiveness-related features, which are typically absent in benign tumors. This broader context enhances the model's ability to distinguish between benign and malignant masses more accurately.

Regarding the suggestion to use segmentation results as additional input (e.g., as a channel), we did consider this during model development but did not observe significant improvements in performance. Considering the imperfect performance of the segmentation model, we opted not to include it directly in the classification process. Instead, we utilized the segmentation results to locate the tumor center and crop the image, ensuring that the model focuses on the region of interest without relying on potentially inaccurate segmentation masks. This method allows the model to learn relevant features from both the tumor and its surrounding context without being misled by potential segmentation errors.

Regarding your observation about the Class Activation Mapping (CAM) in Figure 2C showing attention to background areas, we would like to reference our response to Reviewer #1 from the first-round review:

“Can you explain the inconsistency in CAM results? What specific about these cases?

Response: Thank you for raising this interesting point. In our manuscript, we’ve employed Class Activation Mapping (CAM) as a tool to enhance the interpretability of our model's decisions, primarily by illustrating the focus areas of the model within the imaging data. The majority of CAM visualizations effectively pinpoint the tumor regions, showcasing the model's capacity to identify relevant anatomical features. However, the intrinsic limitations of CAM, notably its coarse spatial resolution stemming from the model's receptive field, preclude detailed pixel-level analysis.

Upon revisiting the AI’s inaccuracies as you suggested, we observed instances where the CAM highlights diverged from the expected tumor locations, instead zeroing in on atypical structures outside the kidneys, such as bone tissues. This deviation may partially be attributed to a tendency towards overfitting, despite our efforts to mitigate this by integrating a relatively straightforward ResNet-18 architecture into our framework. Your observation serves as a critical reminder of the importance of continuing to delve into the realms of AI interpretability and the need for rigorous quality control in AI-driven research. This insight will undoubtedly shape our future investigations.”

5. For the feature extractor ResNet18, the authors have never mentioned if they trained

or finetuned ResNet18 using CT images, so I suppose they directly used the ResNet18 trained on ImageNet. In this case, how to ensure the ResNet18 trained using natural images could generalize well to medical images and particularly extract representative features. Why not use the nnUnet trained using CT images for segmentation to extract features? What if the multi-scale features extracted by the nnUnet from cropped 3D volumes are used for binary classification? In Table s3, all backbones were trained on ImageNet, but tested on CT slices. Due to the distribution shift between training and test, it is unfair to compare them. The authors should also include backbones which have met CT images for comparison.

Response: Thank you for your comment. Firstly, we apologize for any confusion regarding the training of ResNet-18. We did **NOT** directly use the ResNet-18 model trained on ImageNet for inference. Instead, we initialized the ResNet-18 weights from the pre-trained ImageNet model and then fine-tuned the entire network on our curated dataset of CT images. This fine-tuning process allows the model to adapt to domain-specific features present in CT images, enhancing its ability to generalize well to medical imaging tasks. By leveraging transfer learning, we benefit from the rich feature representations learned from ImageNet while tailoring the model to our specific application. Regarding your concern about the models in Table S3, all backbone architectures listed were treated consistently. Each backbone was initialized with ImageNet pre-trained weights and then fine-tuned on the same CT dataset used in our study. This uniform approach ensures a fair comparison among different architectures, as they all underwent the same training process on identical CT data, thus mitigating any distribution shift between training and testing phases.

We have revised the Methods section of the manuscript to clearly state that ResNet-18 and all other backbone architectures were fine-tuned on our CT dataset after being initialized with ImageNet pre-trained weights. This clarification ensures that readers understand our training process and the steps taken to adapt the models to medical imaging data.

6.1. AUC cannot completely represent the classification performance, especially when samples are imbalanced, which however is the case for this work. The authors also need to report AUPRC, which is much better for imbalanced data.

Response: Thank you for your insightful comment. We acknowledge the issue of class imbalance in our dataset and agree that AUC alone may not fully represent the classification performance. To align with clinical practice, we did compare AI model's performance to that of radiologists at different specificity levels:

"Specifically, when the model matched the average sensitivity of these radiologists, its specificity surpassed the average specificity of them. Conversely, when it matched their average specificity, its sensitivity outperformed the average sensitivity of them (Figure 2K). We than compared the sensitivities of the AI system and the reader performance at specificities fixed to match reader performance, and found that the deep learning model significantly outperform 4 radiologists (Table S5). When the sensitivities were matched, the deep learning model demonstrated higher specificity than 6 radiologists but none of them reached a statistical significance (Table S5)."

In response to your suggestion, we have added the AUPRC (Area Under the Precision-Recall Curve) metric in Table S4 and Table S8, which is more appropriate for evaluating performance with imbalanced data.

6.2. In table S4, although the classification model achieves comparable performance in terms of AUC, but significantly lower accuracy and sensitivity scores for TCIA, demonstrating higher false negative rate during diagnosis and leading to loss of effective treatments for patients. One reason may be the portion of benign cases in TCIA is only 3.6%, which is greatly smaller than that of training set (16.8%).

Response: Thank you for your insightful comment. As you noted, the predictive accuracy of our model in distinguishing benign from malignant renal masses in the TCIA cohort was lower due to significantly different inclusion criteria. The TCIA cohort comprised the TCGA-KIRC, TCGA-KIRP, TCGA-KICH, CPTAC-CCRCC, and C4KC-KiTS datasets. Notably, only the C4KC-KiTS dataset contained benign lesions, thus the percentage of benign lesions was significantly lower. This substantial class imbalance affects the model's performance metrics, particularly accuracy and sensitivity, as the model encounters far fewer benign cases in the TCIA dataset. Moreover, the TCIA cohort may not be the most suitable choice for evaluating the model's ability to differentiate between benign and malignant renal masses due to this low prevalence of benign cases. Our primary objective in validating the model on the TCIA dataset was to assess its ability to distinguish between aggressive and indolent renal masses, especially in a cohort with missing CT phases, and to explore potential underlying biological mechanisms.

6.3 However, my main concern is about the race bias of this AI model. The patients of training set completely come from China, while most of patients in TCIA are white. The authors should include the statistics of race in table 1 and report the classification performance for each race group to validate if their models exist the issue of race bias.

Response: Thank you for your thoughtful comment regarding the potential for race bias in our AI model. We acknowledge that our training set consists entirely of patients from China, which is a limitation we have highlighted in the revised manuscript. Accessing diverse, multi-racial datasets is indeed challenging, but we recognize the importance of validating our model across different racial groups.

Regarding race information in the TCIA cohort, it was available for certain patients within the TCGA-KIRC, TCGA-KIRP, and TCGA-KICH cohorts, which include malignant tumors only. Specifically, these cohorts contain 161 White patients, 14 Black or African American patients, and 3 Asian patients. We conducted a subgroup analysis of the White patients in the TCIA cohort and found that our model achieved comparable performance in predicting tumor aggressiveness for this group. Specifically, the model achieved an AUC of 0.726 in White patients from the TCIA cohort. This result indicates that the model performs consistently in this subgroup. We have added these data to the discussion section of the revised manuscript. Unfortunately, the number of Black or African American and Asian patients was too small to provide statistically meaningful results for those subgroups. We acknowledge that a larger and more racially diverse dataset is necessary to thoroughly evaluate potential race bias in our model.

Furthermore, current literature does not suggest significant differences in the biological behavior of renal tumors across different racial groups. For instance, a large-scale study involving 8,421 renal cancer patients demonstrated no significant survival differences between different races [Ref 1]. While this provides some assurance regarding the generalizability of our model, we recognize the need for future studies that include more racially diverse cohorts to fully validate the absence of race bias.

Reference:

[1] Jivanji, Dhaval, et al. "The association between race and 5-year survival in patients with clear cell renal cell carcinoma: a Cohort Study." Urology (2021).

Minor comments:

1. Line 107-109, supplementary material, for patients without arterial phase images in TCIA dataset, how to achieve the registration to the arterial phase image?

Response: Thank you for your comment. This question was addressed in our first-round review, but it may not have been visible in the second-round context. To clarify, for the majority of patients in our study, we had access to complete imaging data, including non-contrast (N), arterial (A), and venous (V) phases. However, in the TCIA dataset, some patients lacked certain phases, such as the arterial phase. In these cases, we manually placed the tumor center on the axial slice with the maximum tumor dimension using the available imaging phases. We have clarified this part in the method section of "Automatic Detection of Renal Masses".

2. For binary classification task, although 2000+ patients were included in training set, the total number of training images are still limited since only one slice was extracted from each CT images. In this case, it is better to train the classifiers by cross-validation.

Response: Thank you for your comment. While cross-validation is often used to evaluate model robustness, we opted not to use it in this study because our primary goal was to train and evaluate a single, well-performing model. Given the nature of deep learning, training five models for cross-validation would have been computationally expensive and time-consuming, especially since our independent test set was sufficiently large and diverse to ensure robust evaluation. Furthermore, cross-validation is typically more beneficial when datasets are small or highly variable, neither of which is the case here. In our case, despite each patient contributing only one slice, the overall dataset was sizable, and the images captured significant variability across different patients and centers. Therefore, we are confident that our training and evaluation approach was appropriate for the data at hand.

3. Line 505, why "survival outcome is the gold standard for aggressiveness evaluation"? Survival outcomes are end points. In fact, Gleason score of biopsy samples is the widely used to evaluate the aggressiveness of cancer.

Response: Thank you for your insightful comment. You are absolutely correct that survival outcomes are endpoints and that pathological assessments, such as the Gleason score in prostate cancer, are widely used to evaluate tumor aggressiveness. In the context of renal tumors, similar pathological indicators include tumor subtype and nuclear grade, which are important for assessing aggressiveness. However, it's important to acknowledge that while these pathological parameters are valuable, they are not always definitive predictors of clinical outcomes. For example, some patients with low tumor nuclear grade may develop distant metastasis, whereas others with high tumor grade may not experience recurrence. This variability suggests that pathological assessments alone may not fully capture the tumor's aggressiveness in terms of its impact on patient survival.

We agree that referring to survival outcome as the "gold standard" for aggressiveness evaluation may not be entirely accurate, as it is an endpoint rather than a diagnostic tool. Survival outcomes provide the most direct measure of the clinical consequences of tumor aggressiveness, but they are not practical for pre-treatment evaluation. We have revised the manuscript to reflect this, acknowledging that survival outcome is an important aspect for evaluating aggressiveness, alongside other pathological indicators.

4. Line 632-634, model overfitting occurs due to limited training samples, not limited test samples. Limited test samples could raise the risk of model generalization as well as model bias.

Response: Thank you for your comment. You are absolutely correct that overfitting is caused by limited training samples, not limited test samples. We apologize for the oversight in our original discussion. We have revised the manuscript to accurately reflect this distinction.

5. Line 649-650, how did you include both tumour nucleus grade and histologic subtype in predicting aggressiveness of renal tumours? Figure shows that only CT images were fed into deep learning models.

Response: Thank you for your insightful comment and for highlighting the need for clarification. We apologize for any confusion caused by our previous statement. To clarify, our deep learning models were trained using only preoperative CT images to predict the aggressiveness of renal tumors. The term "aggressiveness" in our study was defined based on comprehensive pathological evaluations after surgical resection, which included tumor nuclear grade, histologic subtype, and other adverse pathological features such as necrosis and sarcomatoid differentiation. These pathological criteria collectively provided a robust reference standard for classifying tumors as aggressive or indolent. Previous studies have typically focused on predicting single pathological features, such as tumor nuclear grade, and often only for specific subtypes like clear cell renal cell carcinoma. In contrast, our study is the first to use a comprehensive set of pathological criteria to define aggressiveness and to train models that predict this multifaceted outcome from CT images alone.

We recognize that the original wording in the manuscript may have led to misunderstanding. To prevent any further confusion, we have revised the relevant sections of the manuscript to explicitly state that while only CT images were fed into the deep learning models, the definition of aggressiveness was based on a comprehensive pathological evaluation.

6. Line 664, an AUC of 0.619 seems low. In general, the AUC of a random classifier is 0.5, so it seems that the diagnostic model cannot distinguish well between indolent and aggressive tumours in this study.

Response: Thank you for your insightful comment. We acknowledge that an AUC of 0.619 indicates a modest discriminative ability. However, we performed a DeLong test to assess the statistical significance of the AUC compared to 0.5 (the performance of a random classifier). The test confirmed that our model's AUC is significantly better than 0.5, with a p-value of 0.018. This result suggests that the model does provide meaningful discrimination between indolent and aggressive tumors, even though the AUC is not as high as other metrics in the study.

7. The information of spatial resolution for CT scans has not been provided.

Response: Thank you for your comment. We have now provided detailed information on the spatial resolution of the CT scans.

Table S12. Spatial resolution of CT scans

	Training (n = 2400)	Internal test (n =598)	External test (n =561)	Prospective test (n =610)	TCIA test (n =388)
Slice thickness					
0.5-1 mm	3 (0.1%)	0 (0.0%)	5 (0.9%)	5 (0.8%)	1 (0.3%)
1-2 mm	160 (6.7%)	35 (5.9%)	40 (7.1%)	65 (10.7%)	20 (5.2%)
2-3 mm	14 (0.6%)	5 (0.8%)	159 (28.3%)	64 (10.5%)	120 (30.9%)
3-4 mm	245 (10.2%)	58 (9.7%)	301 (53.7%)	212 (34.8%)	44 (11.3%)
4-6 mm	1955 (81.5%)	494 (82.6%)	56 (10.0%)	264 (43.3%)	192 (49.5%)
6-10 mm	23 (1.0%)	6 (1.0%)	0 (0.0%)	0 (0.0%)	11 (2.8%)
In-plane pixel size					
0.4-0.7 mm	1237 (51.5%)	317 (53.0%)	444 (79.1%)	251 (41.1%)	53 (13.7%)
0.7-1.0 mm	1163 (48.5%)	281 (47.0%)	117 (20.9%)	359 (58.9%)	335 (86.3%)

Reviewer #4 (Remarks on code availability):

The authors used the source code of nnUnet for kidney and tumor segmentation, which is reproducible. They used pretrained ResNet on ImageNet to extract features, which is reproducible. Besides, they provided the source code of classification, which is reproducible. Overall, the provided README file is readable, and the source code is a usable resource for the community.

Response: Thank you for taking the time to check the code. We appreciate your encouragement and will continue to improve for the community.

Reviewer #5 (Remarks to the Author):

Introduction: Acceptable

Response: Thank you for your comments.

Methods:

- agree with authors to classify histology as defined rather than based on stage and size.

Response: Thank you for your comments.

- for definition of indolent lesions under histological section, did the authors mean to say invasion instead of evasion? - "included indolent ccRCC (ccRCC without evasion into major veins or perinephric tissues, grade 3-4 components, necrosis or sarcomatoid differentiation), indolent papillary RCC (papillary RCC without evasion into major veins or perinephric tissues, grade 3-4 components or sarcomatoid differentiation)"

Response: Thank you for your suggestions. We agree that 'invasion' is the correct expression and have changed the word 'evasion' to 'invasion'.

- adding the subset analysis for T1/T1a tumors is acceptable

Response: Thank you for your comments.

- agree with authors regarding difficulty to control for size having more aggressive features

Response: Thank you for your comments.

- added section on radiological interpretation is acceptable. I would also add if the 7 readers/expert had any fellowship training in abdominal imaging as part of their 5 years of experience. i.e. Each were fellowship trained in abdominal imaging with over 5 years of experience…

Response: Thank you for your suggestions. The fellowship training was part of the 5 years of experience. We have added this information in the revised manuscript.

- No other concerns in methods section

Response: Thank you for your comments.

Results:

Line 436 - change than to then

Response: Thanks for your correction. We have corrected the word “than” with “then”.

Discussion:

Line 734- typo for "actual"

Table S5 Title - typo errors "radiologists" and "fixed"

Table s6 typo: AUCs for deep "learnig" model x 2

Response: Thanks for your correction. We have corrected all the typos.

Overall summary: Excellent manuscript for which the authors adequately addressed the reviewer's concerns.

Response: Thank you for your positive feedback and recognition of our manuscript. We appreciate your constructive suggestions and are glad to have adequately addressed your concerns.

Reviewer #5 (Remarks on code availability):

NA

REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

Thank you for addressing the concerns, I find the revision to be satisfactory.

Reviewer #4 (Remarks to the Author):

My concerns have been well addressed, particularly on the potential race bias of AI models. Only some minor comments. Some texts in Figure S1 have red underlines due to language detection tools, which should be avoided. The usage of curved brackets and square brackets are inconsistent in Table S4 and S8.

Reviewer #4 (Remarks on code availability):

The authors have provided pretrained model weights, which makes the code more applicable.

Point by point responses to reviewers

Reviewer #1 (Remarks to the Author):

Thank you for addressing the concerns, I find the revision to be satisfactory.

Thanks for your comments.

Reviewer #4 (Remarks to the Author):

My concerns have been well addressed, particularly on the potential race bias of AI models. Only some minor comments. Some texts in Figure S1 have red underlines due to language detection tools, which should be avoided. The usage of curved brackets and square brackets are inconsistent in Table S4 and S8.

Thanks for your comments. We replaced Figure S1 so that texts with red underlines due to language detection tools were avoided. We also used consistent curved brackets and square brackets in Table S4 and S8 in the revised manuscript.

Reviewer #4 (Remarks on code availability):

The authors have provided pretrained model weights, which makes the code more applicable.

Thanks for your comments.